

REAL-TIME PHISHING EMAIL ANALYSIS & INCIDENT RESPONSE

Strong Evidence Portfolio for GitHub & Interview Presentation

Project Type	SOC / Email Security / Incident Response
Environment	Mozilla Thunderbird, Gmail, PowerShell, VirusTotal
Method	Controlled phishing-awareness simulation
Core Evidence	Email creation and delivery, attachment analysis, SHA-256 workflow, threat-intelligence validation, analyst verdict
Analyst	Saikumar

Project Objective

Demonstrate an end-to-end phishing investigation workflow using a safe, self-generated email simulation. The project focuses on identifying suspicious indicators, examining message evidence, validating URLs/files with threat-intelligence tools, documenting IOCs, and producing a defensible analyst verdict.

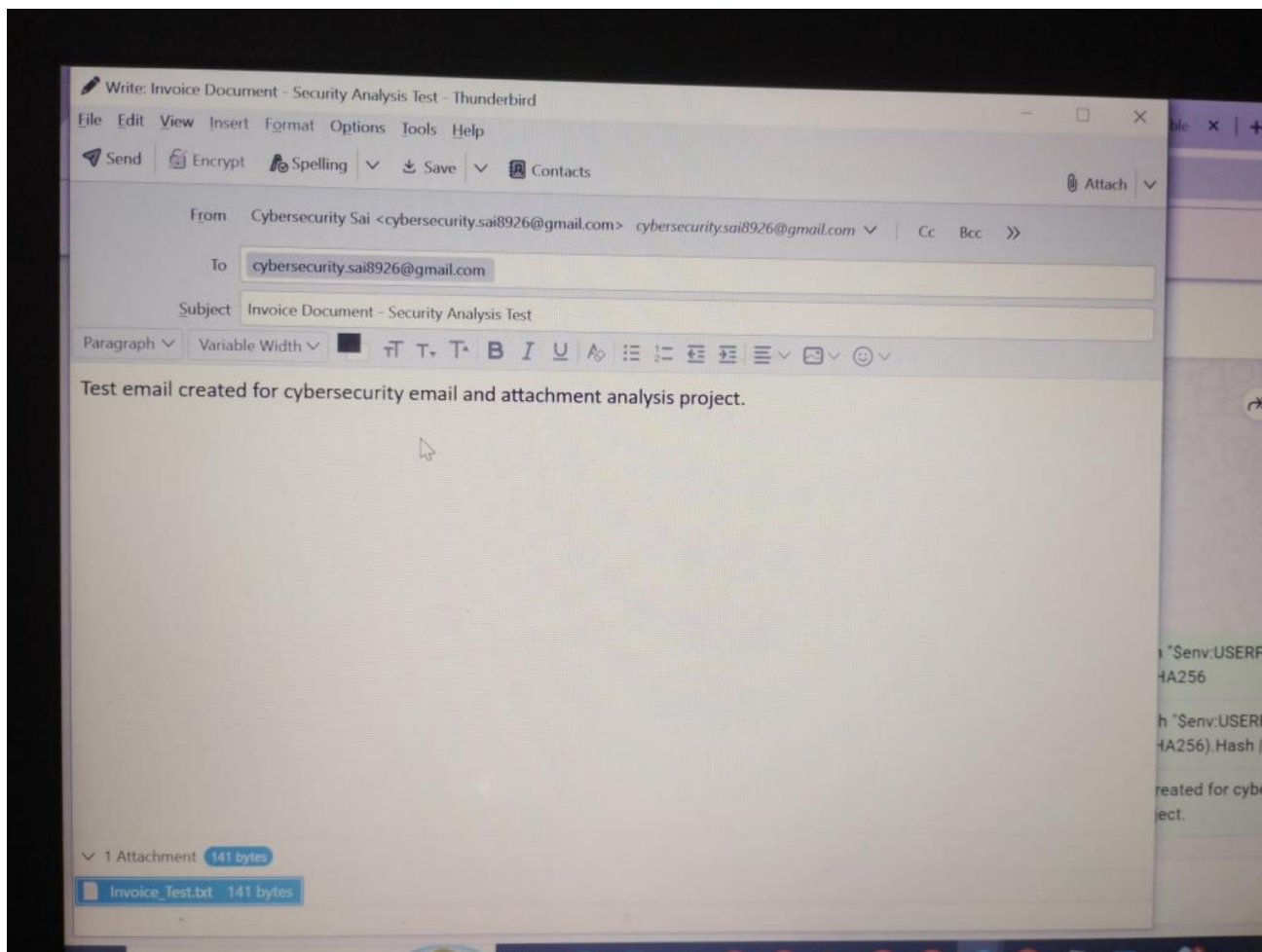
Safety note: This evidence documents a controlled cybersecurity training exercise. No credential harvesting, malicious payload, or real victim targeting was used.

1. Evidence Overview

The strongest evidence below was selected from the completed lab workflow. Together, the screenshots show that the project was not only theoretical: a test email and attachment were created, delivered, investigated, submitted for reputation analysis, and documented in an analyst report.

Evidence	What it proves
Controlled email composition	A test message and attachment were deliberately created for the lab.
Inbox delivery	The phishing-style message and attachment reached the analysis mailbox.
VirusTotal submission	The attachment was submitted for independent reputation / malware checks.
VirusTotal result	The test attachment returned 0/59 detections at analysis time.
Investigation report	Suspicious indicators, header review, URL analysis, attachment analysis, and final classification were documented.
Evidence folder	The project retained a structured screenshot set including email, headers, DMARC, URL, hash, and attachment evidence.

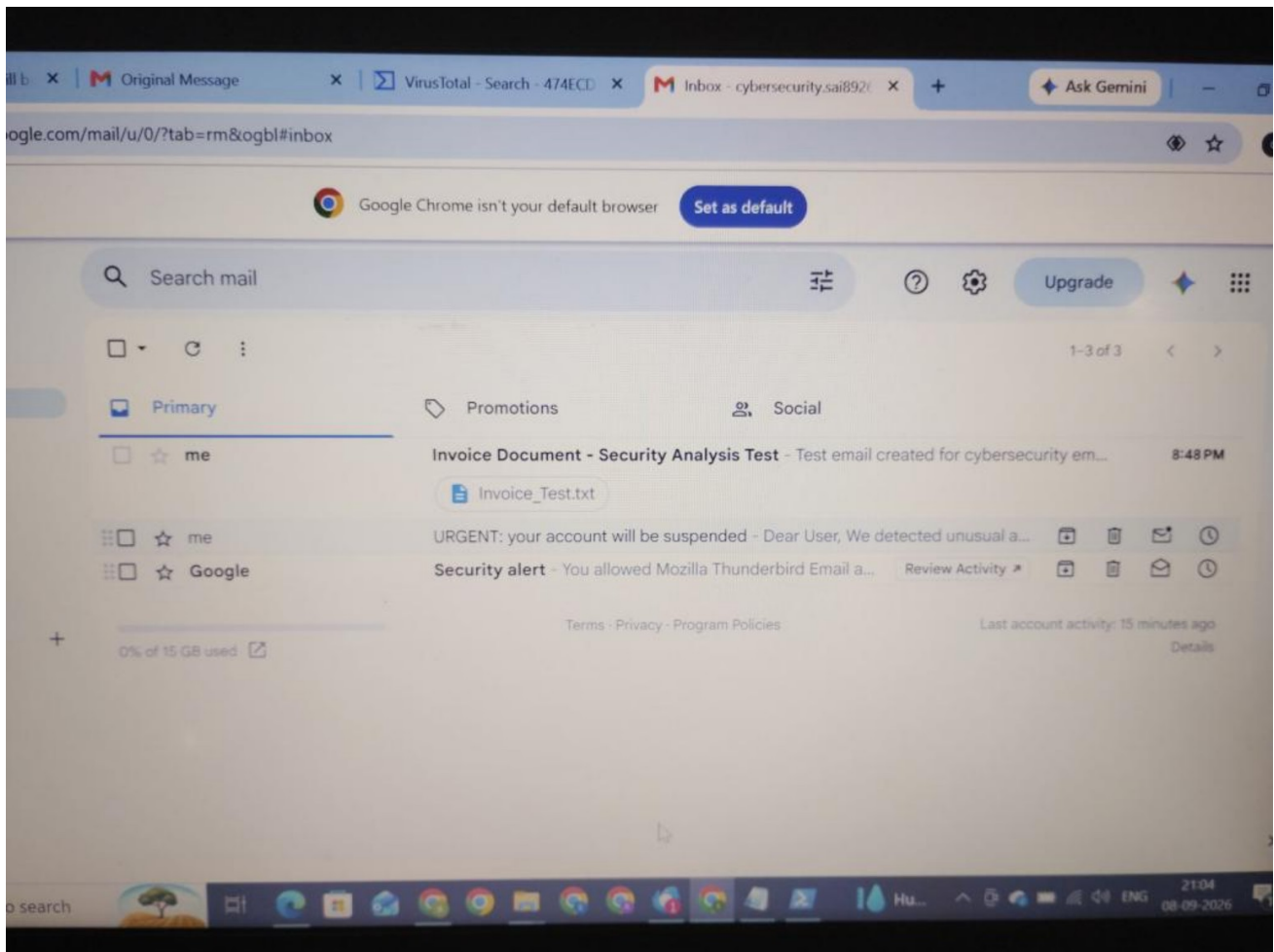
2. Controlled Email & Attachment Creation



The Thunderbird compose window shows a deliberately created test email titled "Invoice Document - Security Analysis Test" with the attachment "Invoice_Test.txt".

Why this is strong evidence: This establishes the controlled origin of the sample and shows that the attachment used later in the investigation was intentionally created for cybersecurity analysis.

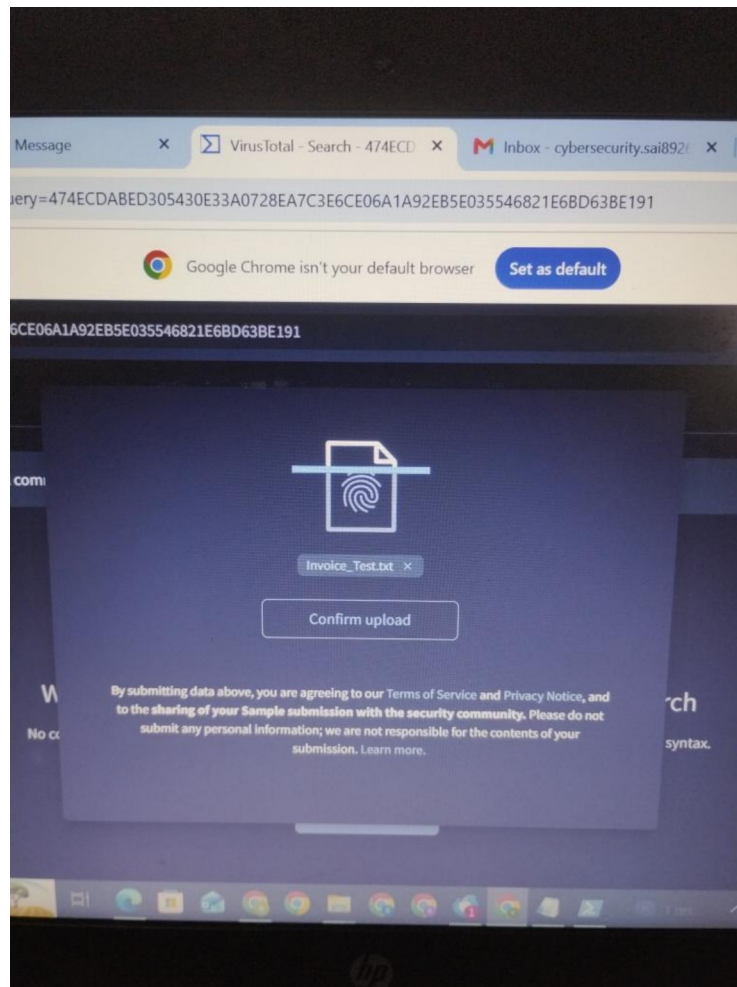
3. Delivery to the Analysis Mailbox



The Gmail inbox shows both the phishing-style message “URGENT: your account will be suspended” and the separate attachment-analysis test message with “Invoice_Test.txt”.

Why this is strong evidence: This demonstrates that the messages were actually delivered and available for analyst triage rather than being described only in notes.

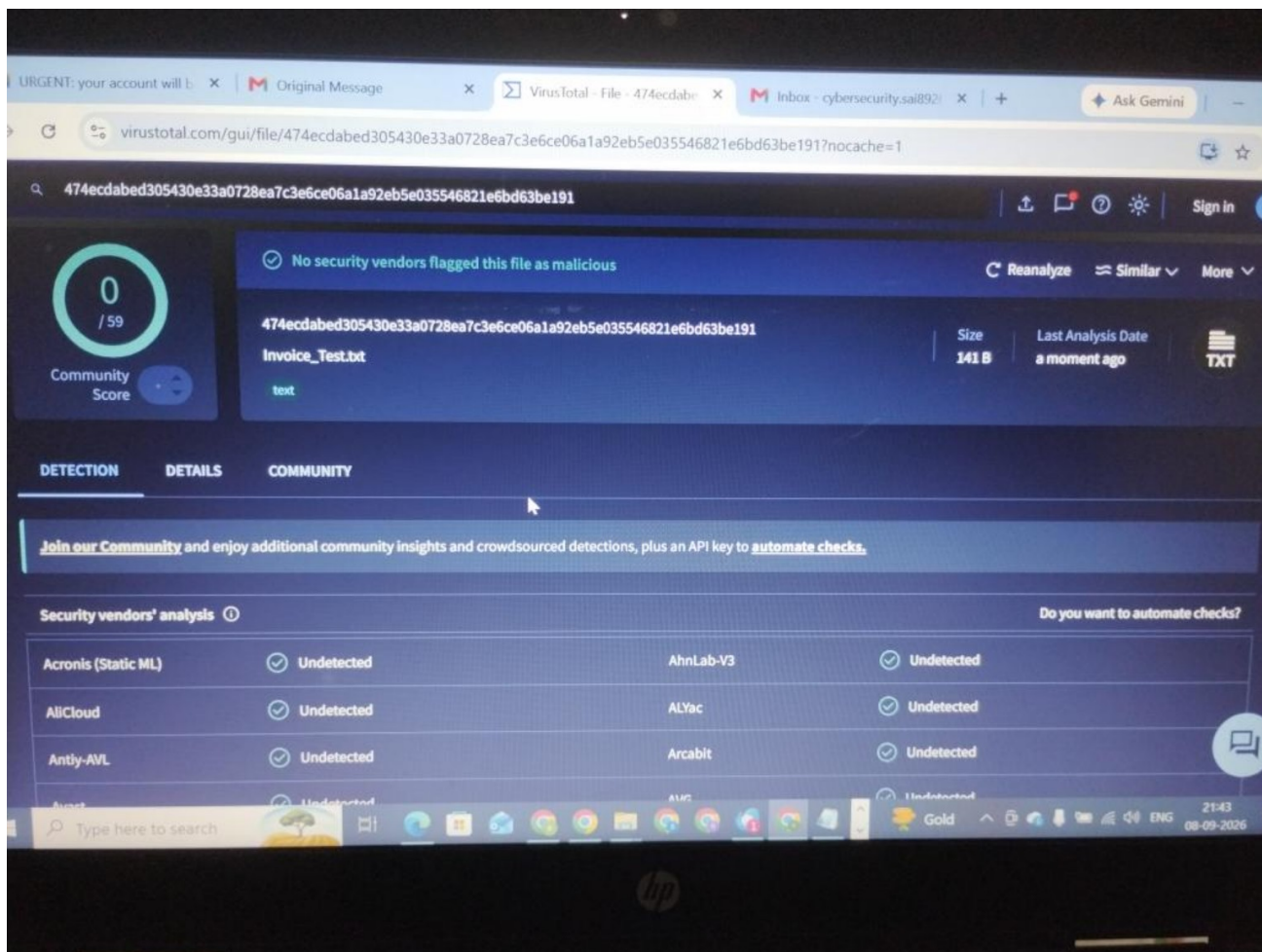
4. Attachment Submission for Threat-Intelligence Analysis



The VirusTotal upload workflow shows “Invoice_Test.txt” selected for analysis.

Why this is strong evidence: This captures the transition from email triage to attachment reputation analysis using an external threat-intelligence platform.

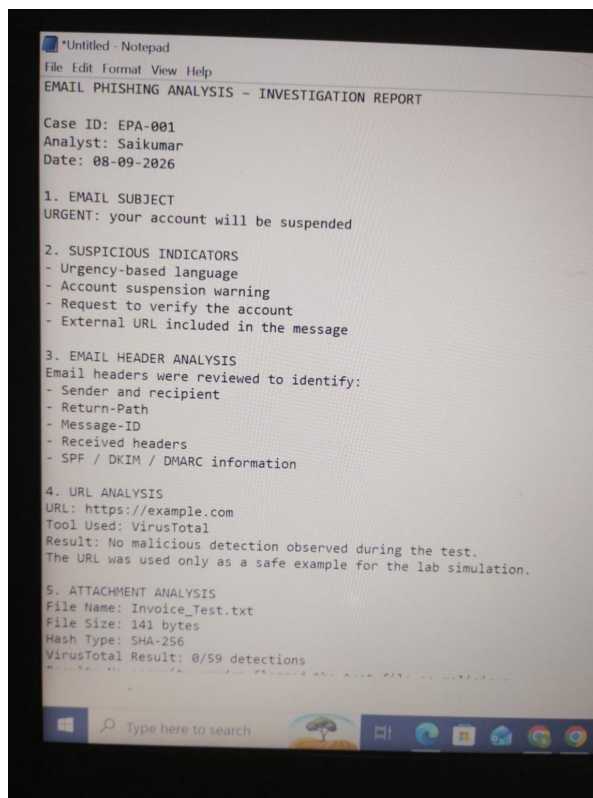
5. VirusTotal Attachment Result



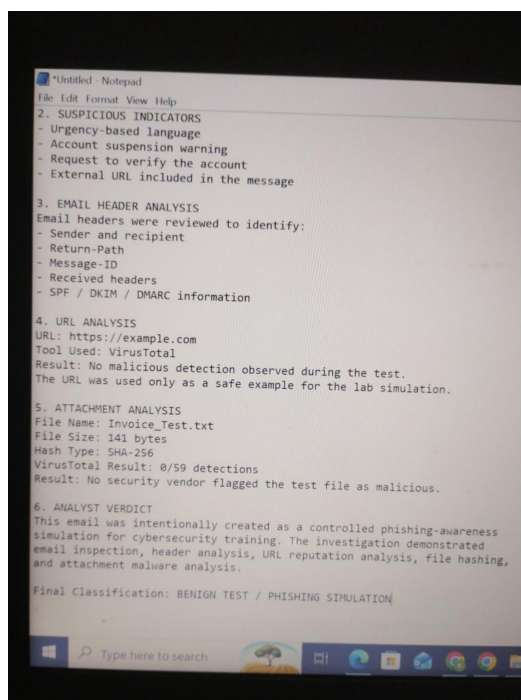
VirusTotal reports that no security vendors flagged "Invoice_Test.txt" as malicious, with a 0/59 detection result visible in the screenshot.

Why this is strong evidence: A clean reputation result does not prove a file is universally safe, but in this controlled lab it supports the conclusion that the attachment was a benign test artifact.

6. Investigation Report - Analysis & Verdict

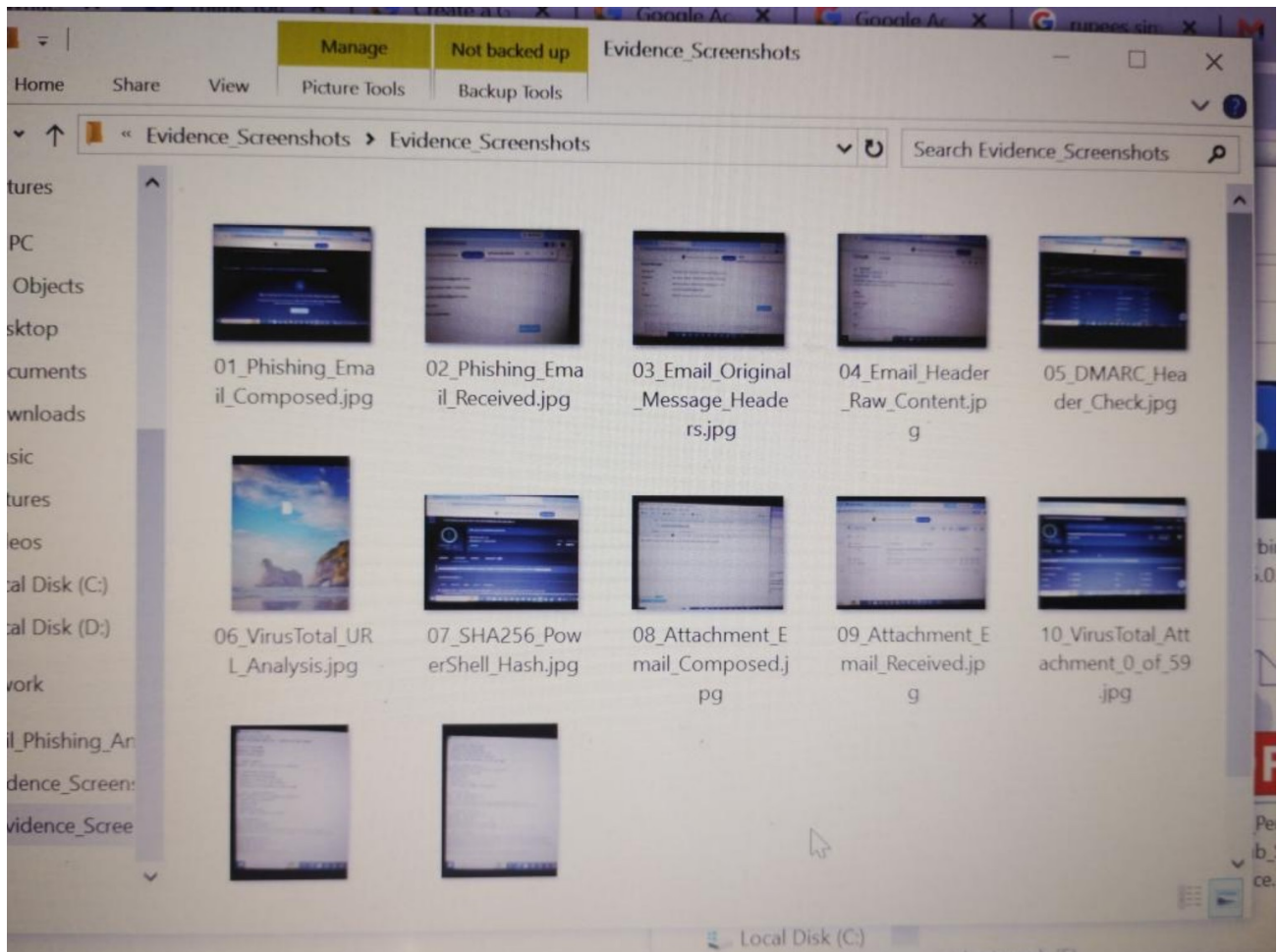


The investigation report records the case ID, phishing-style subject, suspicious indicators, email-header review items, URL analysis, attachment analysis, and VirusTotal result.



The final analyst verdict classifies the exercise as a benign test / phishing simulation and documents the reasoning behind that decision.

7. Structured Evidence Retention



The retained evidence folder contains named screenshots for phishing email composition/receipt, original headers, raw header content, DMARC checking, VirusTotal URL analysis, SHA-256 hashing, attachment email evidence, and VirusTotal attachment analysis.

Why this matters: A professional security investigation should preserve evidence in a clear, reviewable structure. This screenshot demonstrates that the project artifacts were organized rather than left as temporary browser activity.

8. Final Assessment

Final classification: BENIGN TEST / PHISHING SIMULATION

The project demonstrates a realistic SOC-style workflow: receive and triage a suspicious message, identify social-engineering indicators, review email-header artifacts, inspect URL/file indicators, generate or verify a SHA-256 hash, consult threat intelligence, preserve evidence, and document an analyst verdict.

Interview-ready skills demonstrated

- Phishing triage and suspicious-indicator identification
- Email-header analysis concepts: sender, Return-Path, Message-ID, Received headers, SPF/DKIM/DMARC
- URL reputation analysis and safe handling of suspicious links
- Attachment analysis and SHA-256 hashing workflow
- VirusTotal threat-intelligence validation
- IOC-oriented evidence collection and documentation
- Incident-response style classification and final analyst reporting

GitHub recommendation: Upload this PDF together with a README explaining the objective, tools, investigation steps, findings, screenshots, lessons learned, and a clear statement that the activity was performed in a controlled lab.